

Homework 1

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STAT651 - Statistical Theory I
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1 Question 1

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> sample( 1:10, 5, replace=TRUE )  
[1] 8 2 2 1 9  
  
> rnorm( 5, mean=1.0, sd=0.5 )  
[1] 1.909263 1.009593 1.101609 1.534538 1.542092
```

2 Question 2

$$I_1 = \int_0^1 x^2 dx = \frac{1}{3} x^3 \Big|_0^1 = \frac{1}{3}$$
$$I_2 = \int_0^1 \int_0^{2y} x dx dy = \int_0^1 \left(0.5 x^2 \Big|_0^{2y} \right) dy$$
$$= \int_0^1 2y^2 dy = \frac{2}{3} y^3 \Big|_0^1 = \frac{2}{3}$$

Wolfram Alpha agrees with me on both counts.

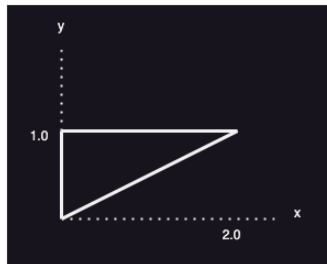


Figure 1: Boundaries of Area Integrated in I_2

3 Question 3

- a Finite - $2^4 = 16$ outcomes. HHHH, THHH, HTHH, TTHH, HHTH, THTH, HTTH, TTTH, HHHT, THHT, HTHT, TTHT, HHTT, THTT, HTTT, TTTT
- b Finite - 4 Heads, 3 Heads 1 Tail, 2 Heads 2 Tails, 1 Head 3 Tails, 4 Tails
- c Countably Infinite - the set I'd choose is the positive integers. While I'm sure there is a maximum number of leaves over 2" in the universe right now, life finds a way and could create a plant with more leaves over 2".
- d Finite - I'm going to assume that lifetime here means lifetime while in use with a specific load. Given a AAA battery has a finite amount of built up charge, even assuming we can use all of it 100% efficiently that'll give us some finite time of use under load. Let's call that value (in hours) M for the lifetime of an ideal AAA (100% efficient) battery. Our set then is $[0, 1, \dots, M]$ and is therefore finite.
- e Uncountably Infinite - We're working with real numbers now so this will be a set of reals that has some range (from a small but positive value as foals can't be flat to some reasonable upper limit).
- f Finite - $S = \{x/50 \mid x \in [0, 1, 2, \dots, 50]\}$